1.47 Sampling Methods

Worked Solutions

1)

Opportunity

2)

Simple random

3)

Systematic

4)

Cluster

5)

Quota

6 a)

The 7200 employees

6 b)

Method B will collect responses from 5 of the 24 stores, so it will less likely be biased based on location. Method C will only consider the London store, and there could be job satisfaction issues based on the location, which is not reflected elsewhere.

6 c)

Method B will collect responses from just 5 stores, but Method A could potentially seek responses from all 24 stores. Method B is therefore more likely to be quicker to implement. Method A also has the potential to collect more responses from some stores than others, introducing potential bias.

7)

A census can be very expensive and time-consuming to both collect and analyse.

8)

There is no sampling frame i.e. we do not have a list of all of the African penguins (the number of breeding pairs is only an estimate, and will regularly fluctuate anyway).

9)

Attendees to a national union conference will mostly be made up of union representatives and so will already have strong views on the conditions in their place of work. So it is unlikely that Chloe will gather many moderate views. She is also relying on attendees visiting her stall and then deciding to fill in the questionnaire, so she will only get the views of people who are willing to do this (even if her stall is close to the entrance, there is no guarantee that everyone will visit it).

10)

Clive should first number all of the customers from 001 to 680. He should then use a random number generator to generate three-digit numbers from 001 to 680. He should generate 40 numbers from 001 to 680, ignoring repeats, and select the customers given those numbers.

11)

$$\frac{3265}{3265+2435} \times 150 = 85.921\ldots \approx 86$$
$$\frac{2435}{3265+2435} \times 150 = 64.078\ldots \approx 64$$

So 86 should be surveyed from the shops and 64 from the warehouses.

12)

A simple random sample would ask the opinion of a number of people who would not be affected and would be a waste of their time. An opportunity sample would be more appropriate as it would gather the opinions of those who work near the kitchen who are the most likely people to own the mugs. It would also be very easy to gain opinions quickly.

13)

Susan should first number all of the classes from 01 to 50. She should then use a random number generator to generate two-digit numbers from 01 to 50. She should generate 3 numbers from 01 to 50, ignoring repeats, and survey all of the students in the classes given those numbers.

14)

Some members of staff may not check their emails within the 24-hour window as they may be ill or only work part-time. Of those that do read the email, the responses may only come from those with a strong interest or motivation to take part, and hence may lead to a biased sample.

15 a)

Total number of people in the choir:

$$25 + 30 + 50 + 45 = 150$$

Aurora:

$$\frac{25}{150} \times k = 8 \Rightarrow k = 48$$

BrightByte:

$$\frac{30}{150} \times k = 10 \Rightarrow k = 50$$

Cosmic:

$$\frac{50}{150} \times k = 17 \Rightarrow k = 51$$

If $k = 48$ then

$$\frac{25}{150} \times 48 = 8, \quad \frac{30}{150} \times 48 = 9.6 \approx 10, \quad \frac{50}{150} \times 48 = 16 \neq 17 \text{ so } k \neq 48$$

If $k = 49$ then

$$\frac{25}{150} \times 49 = 8.16\ldots \approx 8, \quad \frac{30}{150} \times 49 = 9.8 \approx 10, \quad \frac{50}{150} \times 49 = 16.33\ldots \neq 17$$

so $k \neq 49$

If $k = 50$ then

$$\frac{25}{150} \times 50 = 8.33\ldots \approx 8, \quad \frac{30}{150} \times 50 = 10, \quad \frac{50}{150} \times 50 = 16.66\ldots \approx 17$$

so k may equal 50

If $k = 51$ then

$$\frac{25}{150} \times 51 = 8.5\ldots \neq 8, \quad \frac{30}{150} \times 51 = 10.2 \approx 10, \quad \frac{50}{150} \times 51 = 17 \text{ so } k \neq 51$$

So $k = 50$

15 b)

Dreamscape:

$$\frac{45}{150} \times 50 = 15$$

So 15 Dreamscape employees should be sampled.

16 a)

Systematic

16 b)

There are only 6 sets of 120 employees that can be selected using this method. For example: $\{1, 7, 13, 19, \ldots\}$, $\{2, 8, 14, 20, \ldots\}$ can be selected, but $\{1, 2, 3, 4, \ldots\}$ cannot. So because not all samples of size 20 can be selected, the sampling method is not random.